

NAKHON SI THAMMARAT, Thailand

WMO: 485520

Lat: 8.5461N

Lon: 99.9396E

Elev: 18

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	70.7	72.1	67.1	99.9	78.6	68.5	104.8	78.9	13.6	81.4	11.9	82.1	0.1	330	0.299

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	15.0	95.0	79.4	93.8	79.2	92.7	79.2	81.6	91.2	80.9	90.1	80.5	89.4	5.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.2	151.4	85.3	78.7	148.9	84.9	78.2	146.5	84.4	45.3	91.6	44.7	90.6	44.1	89.7	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.4	9.7	8.3	DB	68.2	97.0	2.0	1.2	66.7	97.9	65.5	98.6	64.4	99.3	62.9	100.2	
			WB	65.7	83.4	2.7	2.2	63.7	85.0	62.1	86.3	60.5	87.5	58.6	89.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.2	79.5	80.4	82.1	84.0	84.6	84.3	83.8	83.6	82.9	81.6	80.2	79.4
	DBStd	2.55	1.65	1.70	1.82	1.98	1.74	1.55	1.61	1.66	1.50	1.82	2.10	1.79
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11760	915	850	996	1021	1073	1028	1049	1041	988	981	905	913
	CDD65	6285	450	430	531	571	608	578	584	576	538	516	455	448
	CDH74	63325	3846	4140	5661	6566	6783	6434	6289	6068	5302	4746	3861	3629
	CDH80	24236	1095	1500	2347	2954	2942	2747	2615	2482	1998	1569	1076	911
Wind	WSAvg	1.7	2.0	2.2	2.2	1.7	1.5	1.5	1.6	1.7	1.3	1.3	1.6	2.1
Precipitation	PrecAvg	99.2	9.3	2.6	2.0	3.4	7.1	3.3	4.3	3.5	6.5	14.4	23.9	18.9
	PrecMax	141.5	48.8	7.0	4.6	9.0	12.5	7.8	7.0	6.7	13.0	25.2	34.4	34.3
	PrecMin	66.0	0.6	0.1	0.0	0.7	1.2	1.3	0.7	0.9	2.4	3.9	13.1	7.7
	PrecStd	19.0	11.9	2.2	1.5	2.8	3.1	1.8	2.0	1.7	3.1	5.4	6.7	8.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.3	91.8	94.1	96.6	96.7	95.3	95.2	95.1	93.6	91.8	89.9	89.0
		MCWB	78.1	78.6	79.5	81.0	80.6	78.6	78.0	77.8	78.4	79.3	79.0	78.3
	2%	DB	87.8	89.7	92.1	94.5	94.5	93.8	93.4	93.2	91.8	90.2	88.6	87.5
		MCWB	77.3	77.7	79.1	80.4	80.2	79.0	78.3	78.0	78.7	79.1	78.8	78.0
	5%	DB	86.6	88.3	90.7	92.9	93.0	92.5	92.0	91.8	90.6	89.0	87.5	86.3
		MCWB	76.9	77.2	78.7	80.0	80.0	79.1	78.4	78.2	78.7	78.8	78.6	77.6
	10%	DB	85.5	87.2	89.4	91.4	91.4	91.1	90.5	90.3	89.1	87.6	86.0	85.0
		MCWB	76.6	76.9	78.3	79.5	79.8	79.1	78.3	78.3	78.5	78.6	78.2	77.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.8	79.8	81.0	82.5	82.4	81.8	80.9	80.9	81.0	80.9	81.0	80.5
		MCDB	86.5	89.1	90.6	93.9	92.7	91.3	90.0	89.9	89.4	88.6	87.2	86.1
	2%	WB	78.9	78.9	80.0	81.4	81.6	80.9	80.0	80.1	80.1	80.0	80.0	79.4
		MCDB	85.3	87.5	89.4	91.7	91.4	90.2	89.1	89.1	88.4	87.3	86.0	85.0
	5%	WB	78.1	78.2	79.4	80.7	80.9	80.2	79.4	79.4	79.4	79.5	79.3	78.7
		MCDB	84.2	86.2	88.3	90.3	90.3	89.2	88.4	88.3	87.6	86.6	85.1	84.1
	10%	WB	77.4	77.5	78.8	80.0	80.2	79.6	78.9	78.8	78.9	79.0	78.7	77.9
		MCDB	83.1	85.1	87.4	89.0	89.0	88.3	87.7	87.6	86.9	85.7	84.3	82.8
Mean Daily Temperature Range	5% DB	MDBR	11.9	14.5	15.2	15.5	15.0	14.7	14.6	14.7	14.1	12.3	10.2	10.4
		MCDBR	13.9	16.1	17.2	17.7	17.0	16.7	16.6	16.8	15.9	14.6	13.3	13.1
		MCWBR	5.2	6.0	6.0	5.7	5.1	4.9	4.7	4.8	4.9	5.0	5.0	5.0
	5% WB	MCDBR	12.2	14.7	15.9	16.4	15.8	15.1	15.1	15.6	14.8	13.3	11.9	11.3
		MCWBR	5.1	5.8	5.9	5.8	5.4	5.1	5.0	5.2	5.2	5.1	5.0	4.9
Clear-Sky Solar Irradiance	taub	0.494	0.500	0.507	0.478	0.454	0.452	0.453	0.447	0.439	0.453	0.476	0.486	0.486
	taud	2.167	2.144	2.160	2.248	2.354	2.391	2.373	2.380	2.392	2.355	2.270	2.211	2.211
	Ebn at Noon	260	263	262	266	267	265	265	271	277	273	264	260	260
	Edh at Noon	48	50	50	45	40	38	39	39	39	40	43	45	45
All-Sky Solar Radiation	RadAvg	1446	1756	1825	1817	1623	1560	1535	1579	1548	1387	1239	1235	1235
	RadStd	138	121	158	110	93	82	93	79	61	115	119	127	127

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (16 neighbors)	+0.29	N/A	N/A	N/A	N/A	+0.41	N/A	N/A	N/A	N/A
	+0.36	+0.49	N/A	+0.50	+0.36	+0.38	N/A	N/A	+133	+135

Nomenclature: See separate page